



Xi'an Jiaotong-Liverpool University

西交利物浦大學

RBE211TC DYNAMIC SYSTEMS

# MODELLING MECHANICAL SYSTEMS

Dr Sze-Hong Teh

E-mail: [Szehong.Teh@xjtlu.edu.cn](mailto:Szehong.Teh@xjtlu.edu.cn)



# Mechanical System Elements

## Basic Mechanical System Elements

- A mechanical dynamic system is typically composed of three fundamental elements:
  - Inertia (Mass)
  - Stiffness (Spring)
  - Damping (Damper)
- These elements represent how the system:
  - stores kinetic energy (inertia)
  - stores potential energy (stiffness)
  - dissipates energy (damping)
- By combining these elements, we can model the dynamic behaviour of many mechanical systems.



# Mechanical System Elements

## Translational Mechanical Elements

- In translational systems, the dynamic variable is displacement  $x(t)$ .
- The three basic translational mechanical elements are:

- Mass (Inertia Element)

$$F(t) = m\ddot{x}(t)$$

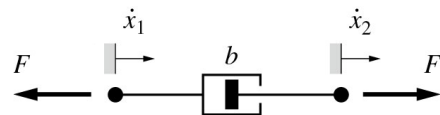
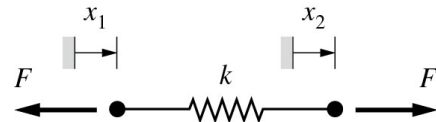
- Spring (Stiffness Element)

$$F(t) = kx(t)$$

- Damper (Damping Element)

$$F(t) = b\dot{x}(t)$$

- These equations describe the force–displacement relationships for each element.



# Mechanical System Elements

## Translational Mechanical Elements: Inertia (Mass)

- The inertia element represents mass in translational motion.
- According to Newton's Second Law:

$$F(t) = m\ddot{x}(t)$$

- This equation relates the applied force to the resulting acceleration.
- The inertia element stores kinetic energy:

$$E_k = \frac{1}{2}m\dot{x}^2(t)$$

- In robotic systems, inertia may arise from links, moving payloads, or actuator components.
- The inertia element introduces a second-order derivative into the system dynamics.



## Translational Mechanical Elements: Stiffness (Spring)

- The stiffness element represents elastic energy storage in translational motion.
- The force generated by a spring is proportional to its deformation:

$$F(t) = kx(t)$$

- The stiffness element stores potential energy:

$$E_p = \frac{1}{2} kx^2(t)$$

- In robotic systems, stiffness may arise from flexible links, joint compliance, or transmission elasticity.

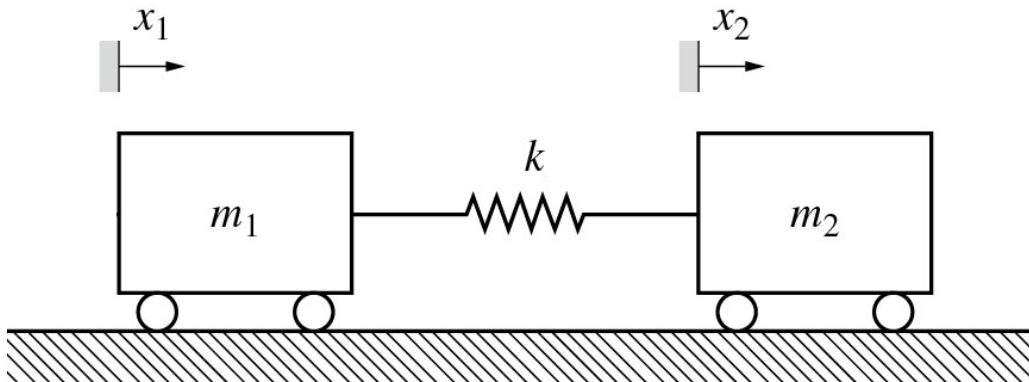


# Mechanical System Elements

## Translational Mechanical Elements: Stiffness (Spring)

- Strictly speaking, spring force depends on the relative displacement between its two ends:

$$F(t) = k(x_1(t) - x_2(t))$$

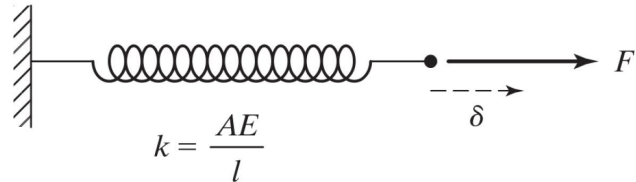
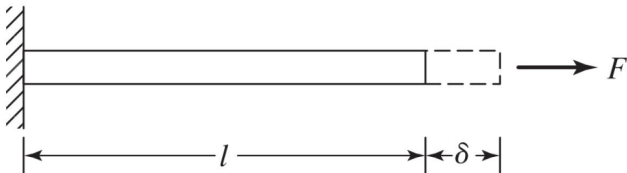


# Mechanical System Elements

## Equivalent Stiffness of Mechanical Elements

- The stiffness coefficient  $k$  depends on material properties and geometry.
- For an elastic rod under axial deformation:

$$k = \frac{EA}{l}$$



Where:

$E$  = Young's modulus

$l$  = length of the element

$A$  = cross-sectional area

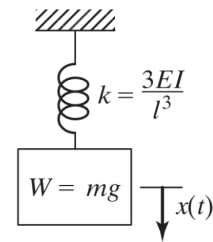
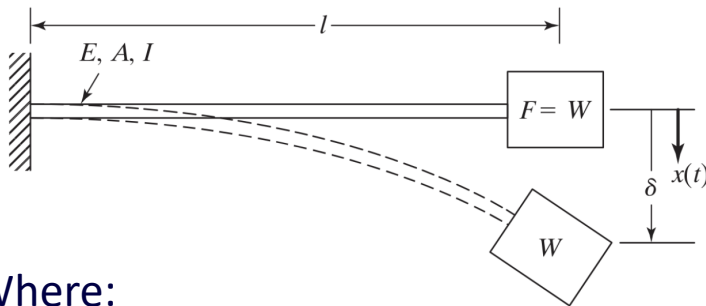


# Mechanical System Elements

## Equivalent Stiffness of Mechanical Elements

- The stiffness coefficient  $k$  depends on material properties and geometry.
- For a cantilever beam under transverse loading:

$$k = \frac{3EI}{l^3}$$



Where:

$E$  = Young's modulus

$I$  = second moment of area

$l$  = length of the element





## Translational Mechanical Elements: Damping (Damper)

- The damping element represents energy dissipation in translational motion.
- The force generated by a viscous damper is proportional to the velocity of motion:

$$F(t) = b\dot{x}(t)$$

- The damping element dissipates mechanical energy as heat.
- In robotic systems, damping may arise from joint friction, viscous fluid resistance, or gear transmission losses.



## Translational Mechanical Elements: Damping (Damper)

- In many practical systems, a damper connects two moving bodies.
- Therefore, the damping force depends on the relative velocity between its two ends:

$$F(t) = b(\dot{x}_1(t) - \dot{x}_2(t))$$

- If both ends move at the same velocity, no damping force is generated.



# Mechanical System Elements

## Rotational Mechanical Elements

- In rotational systems, the dynamic variable is angular displacement  $\theta(t)$ .
- The three basic rotational mechanical elements are:

- Rotational Inertia (Inertia Element)

$$\tau(t) = J\ddot{\theta}(t)$$

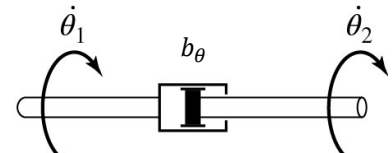
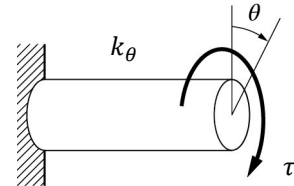
- Torsional Stiffness (Stiffness Element)

$$\tau(t) = k_{\theta}\theta(t)$$

- Rotational Damping (Damping Element)

$$\tau(t) = b_{\theta}\dot{\theta}(t)$$

- These equations describe the torque–angular displacement relationships for each element.



## Rotational Mechanical Elements: Torsional Stiffness

- The stiffness element represents elastic energy storage in rotational motion.
- The torque generated by a spring is proportional to its angular deformation:

$$\tau(t) = k_{\theta}\theta(t)$$

- The stiffness element stores potential energy:

$$E_p = \frac{1}{2} k_{\theta} \theta^2(t)$$

- In robotic systems, stiffness may arise from flexible shafts, joint compliance, or transmission elasticity.

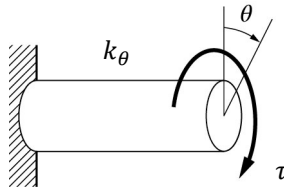


# Mechanical System Elements

## Equivalent Torsional Stiffness of Mechanical Elements

- The stiffness coefficient  $k$  depends on material properties and geometry.
- For a circular shaft under torsion:

$$k_{\theta} = \frac{GJ}{l}$$



Where:

$G$  = shear modulus

$l$  = length of the shaft

$J$  = polar second moment of area



## Rotational Mechanical Elements: Damping

- The damping element represents energy dissipation in rotational motion.
- The torque generated by a viscous rotational damper is proportional to angular velocity:

$$\tau(t) = b_{\theta} \dot{\theta}(t)$$

- The damping element dissipates mechanical energy, usually in the form of heat.
- In robotic systems, damping may arise from joint friction, viscous resistance in actuators, or gear transmission losses.



# Mechanical System Elements

## Lever Mechanism

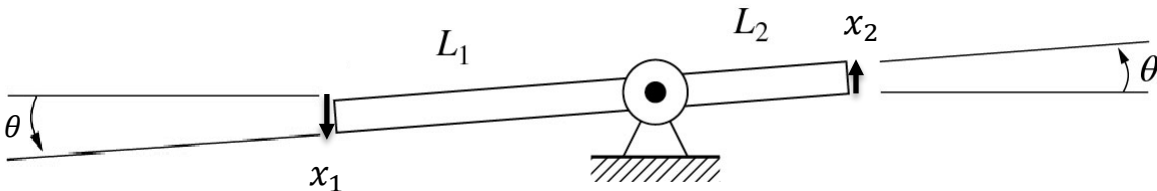
- Rotational motion may be converted into translational motion using a lever.
- From geometry:

$$x_1(t) = L_1 \theta(t) \quad x_2(t) = L_2 \theta(t)$$

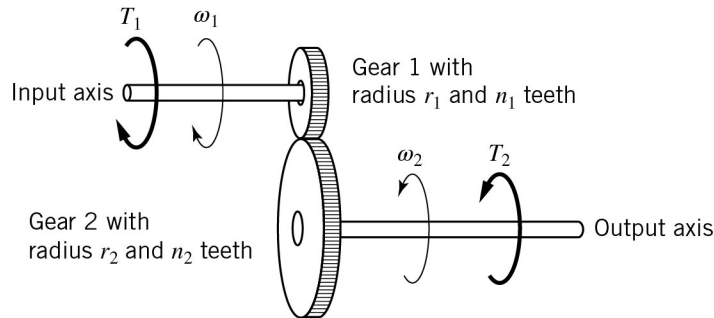
$$\dot{x}_1(t) = L_1 \dot{\theta}(t) \quad \dot{x}_2(t) = L_2 \dot{\theta}(t)$$

$$\frac{x_1(t)}{x_2(t)} = \frac{L_1}{L_2}$$

Where  $L_1$  and  $L_2$  are the distance from the pivot to each end.



## Gear Train



- Rotational motion may also be transformed between shafts using gears.

- For gear radii  $r_1, r_2$ , the gear ratio:

$$\frac{r_2}{r_1} = \frac{n_2}{n_1} = N$$

- The velocity of the mesh point is the same for both gears i.e.

$$V_{mesh} = r_1\omega_1 = r_2\omega_2, \text{ therefore}$$

$$\frac{\omega_1}{\omega_2} = \frac{\theta_1}{\theta_2} = \frac{r_2}{r_1} = N$$



## Mechanical Transformer

- Lever and gear mechanisms act as mechanical transformers.
- They relate motion variables between mechanical domains:

$$x(t) = r\theta(t)$$
$$\theta_1(t) = N\theta_2(t)$$

- This allows translational and rotational elements to be connected in dynamic models.



# Two-Step Modelling Process

## Two-Step Modelling Process

### Step 1 – Physical Modelling

- **Identify mechanical elements**
- Define coordinates and reference directions
- Draw the free-body diagram

### Step 2 – Mathematical Modelling

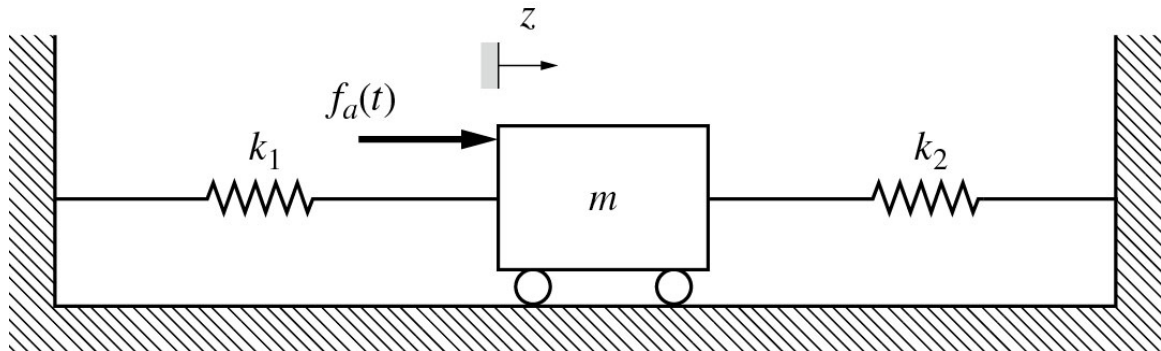
- Apply Newton's 2nd Law (or rotational equivalent)
- Express forces/torques in terms of motion variables
- Obtain governing differential equations



# Mathematical Modelling of Mechanical Systems

## Example 1

Figure below shows a single-mass translational mechanical system. Both springs are undeflected when  $z = 0$  and  $f_a(t) = 0$ . Derive the mathematical model of the mechanical system.



## Example 1

Free-body diagram:



## Example 1

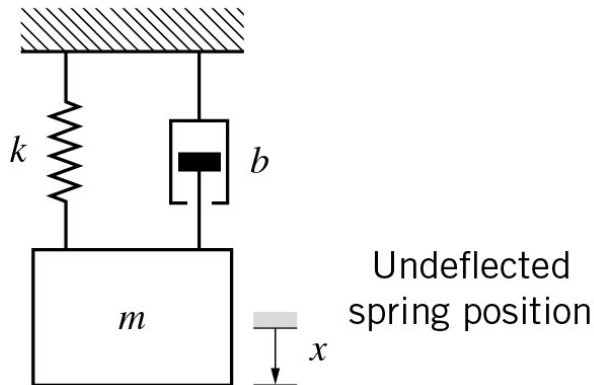
Apply Newton's Second Law:



# Mathematical Modelling of Mechanical Systems

## Example 2

Figure below shows a mechanical system composed of a single mass  $m$ , a spring  $k$ , and a damper  $b$ . The vertical position of the mass is  $x$ , which is measured from the undeflected position of the spring. Derive the mathematical model of the mechanical system. Take downward as positive direction.



## Example 2

Free-body diagram:



## Example 2

Apply Newton's Second Law:





## Example 2

At static equilibrium, velocity and acceleration are zero.

$$\ddot{x} = 0, \quad \dot{x} = 0$$

Substitute:

$$kx_s = mg$$

Therefore:

$$x_s = \frac{mg}{k}$$



## Example 2

Define  $z(t)$  as the position of the mass relative to its static deflection, the total deflection is:

$$x(t) = x_s + z(t)$$



## Example 2

Since:

$$kx_s = mg$$

Therefore:

The above equation is the mathematical model of the system in terms of position variable  $z$  (deflection from static equilibrium).

Hence, gravity shifts only the static equilibrium position and does not influence the dynamic behaviour about equilibrium.



## General SDOF Linear System

- After modelling about equilibrium, a single-degree-of-freedom (SDOF) linear mechanical system takes the form:

$$m\ddot{q}(t) + b\dot{q}(t) + kq(t) = f(t)$$

Where:

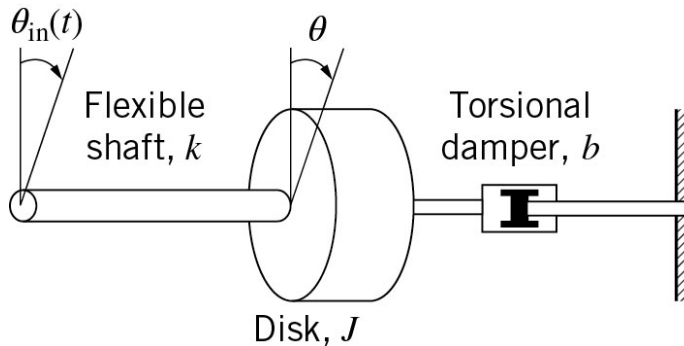
- $m$  = mass (inertia)
  - $b$  = damping coefficient
  - $k$  = stiffness
  - $q(t)$  = displacement about equilibrium
  - $f(t)$  = external force input
- 
- Mechanical systems differ physically, but their mathematical structure is universal.



# Mathematical Modelling of Mechanical Systems

## Example 3

Figure below shows a rotational mechanical system driven by the displacement of the left end of the flexible shaft,  $\theta_{in}(t)$ . When displacements  $\theta_{in}(t) = 0$  and  $\theta = 0$ , the torsional spring  $k$  is not twisted. Derive the mathematical model of the mechanical system.



## Example 3

The free-body diagram of disk  $J$  is shown below, assuming  $\theta_{in}(t) > \theta$ :



## Example 3

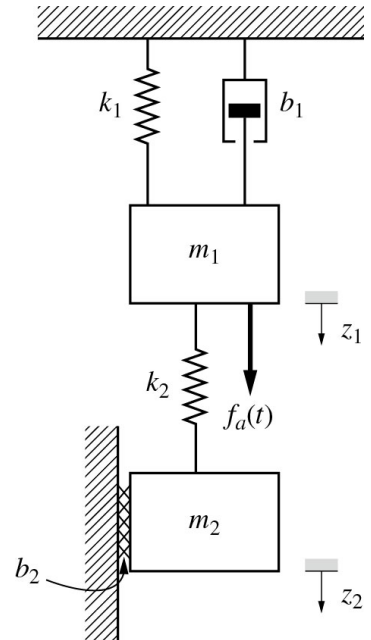
Apply Newton's Second Law for rotation:



# Mathematical Modelling of Mechanical Systems

## Example 4

The right figure shows a vertical mechanical system. All displacements are measured relative to their static equilibrium positions when the applied forces  $f_a(t) = 0$ . Derive the mathematical model of this system.



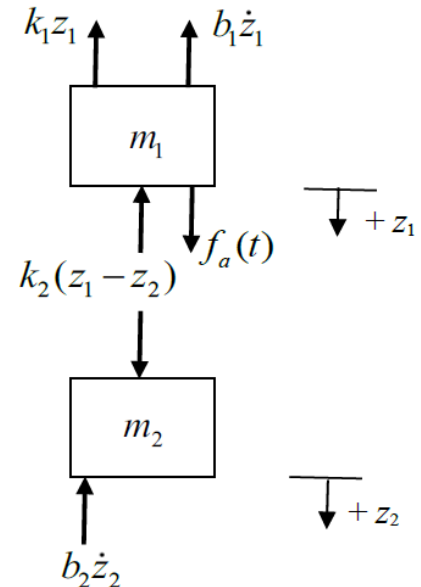


# Mathematical Modelling of Mechanical Systems

## Example 4

The free-body diagrams of  $m_1$  and  $m_2$  are shown below, assuming  $\dot{z}_1 > 0$ ,  $\dot{z}_2 > 0$ , and  $z_1 > z_2$ .

There are no gravity forces in the FBDs because positions  $z_1$  and  $z_2$  are measured from static equilibrium with zero external force.



## Example 4

Applying Newton's Second Law:



## Example 4

Rearrange and put all dynamic variables on the left-hand side and input variables on the right-hand side:

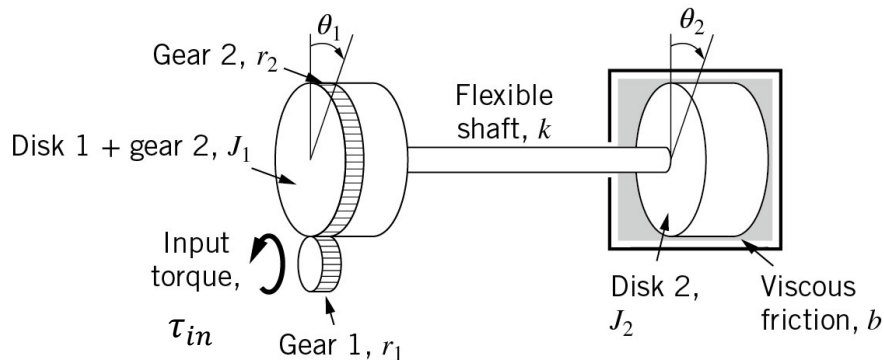
This system can be written compactly in matrix form as:



# Mathematical Modelling of Mechanical Systems

## Example 5

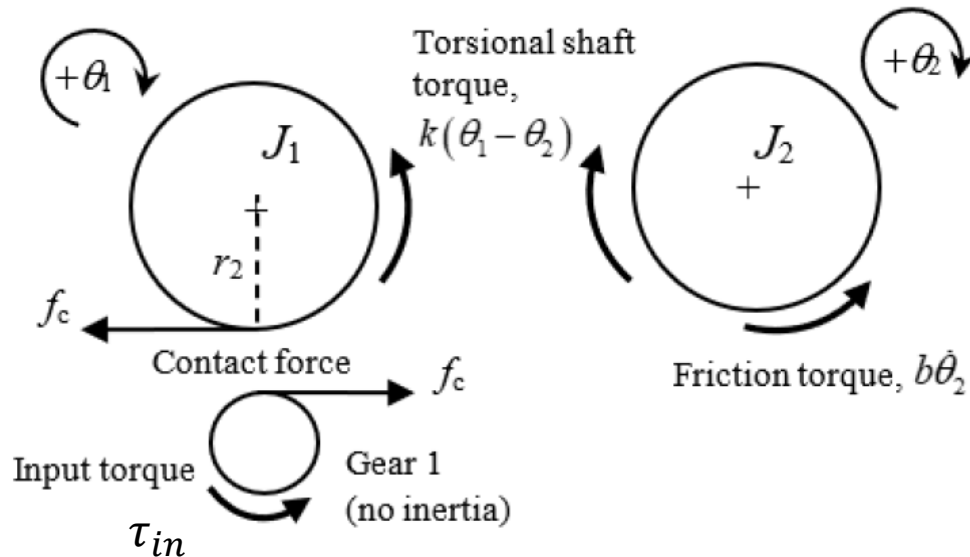
A two-disk rotational mechanical system is shown in figure below. An external motor (not shown) delivers the input torque  $\tau_{in}$  to the input gear 1 (radius  $r_1$ ). The moment of inertia of disk 1 and gear 2 is  $J_1$  and  $J_2$  is the moment of inertia of the second disk. The gear train is ideal. The inertia of gear 1 can be neglected, and gear 2 is larger than gear 1 ( $r_2 > r_1$ ). The two disks are connected by a flexible shaft with torsional spring constant  $k$ . Disk  $J_2$  is immersed in a viscous fluid modeled by friction coefficient  $b$ . Derive the mathematical model of the system.



# Mathematical Modelling of Mechanical Systems

## Example 5

The free-body diagrams of the system, assuming  $\theta_1 > \theta_2$  and  $\dot{\theta}_2 > 0$ :



## Example 5

Apply Newton's Second Law to all three disks:

Gear 1

$$\curvearrowright^+ : \sum \tau_{\theta_{in}} = 0$$

$$\tau_{in} - f_c r_1 = 0$$

Disk 2

$$\curvearrowright^+ : \sum \tau_{\theta_2} = J_2 \ddot{\theta}_2$$

$$k(\theta_1 - \theta_2) - b\dot{\theta}_2 = J_2 \ddot{\theta}_2$$

Disk 1 + Gear 2

$$\curvearrowright^+ : \sum \tau_{\theta_1} = J_1 \ddot{\theta}_1$$

$$f_c r_2 - k(\theta_1 - \theta_2) = J_1 \ddot{\theta}_1$$



## Example 5

Because Gear 1 has negligible inertia,  $\tau_{in} = f_c r_1$  or  $f_c = \frac{1}{r_1} \tau_{in}$ .

Rearrange we obtain:

$$\begin{aligned} \theta_1: & J_1 \ddot{\theta}_1 + k(\theta_1 - \theta_2) = \frac{r_2}{r_1} \tau_{in} \\ \theta_2: & J_2 \ddot{\theta}_2 + b \dot{\theta}_2 + k(\theta_2 - \theta_1) = 0 \end{aligned}$$

This system can be written compactly in matrix form as:



# Review

## 1. Mechanical System Elements

- Inertia
- Stiffness
- Damping

## 2. Two-Step Modelling Process

- Physical modelling
- Mathematical modelling

## 3. Mathematical Modelling of Mechanical Systems

- SDOF systems (translational & rotational)
- MDOF coupled systems
- Geared rotational system
- Matrix representation

